

Name _____

Date _____

Total
/40

KNOWLEDGE

1. Evaluate the integral by interpreting it in terms of area. Include a diagram.

$$\int_{-3}^0 (1 + \sqrt{9 - x^2}) dx$$

(3)

2. Evaluate each of the following.

a. $\frac{d}{dx} \int_3^x e^{t^2-t} dt$

b. $\frac{d}{dx} \int_{3x^2}^7 \sqrt{3t^4 + t - 8} dt$

(1)
(2)

c. $\int_1^9 \frac{x-1}{\sqrt{x}} dx$

d. $\int_1^4 (5 - 2t + 3t^2) dt$

(2)
(2)

e. $\int_0^\pi (1 + \cos x) dx$

f. $\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \csc x \cot x dx$

(2)
(2)

3. Find $h'(x)$ if $h(x) = \int_{5x^4}^{\sec x} \sqrt{4t - 9} \, dt$. Do NOT fully simplify.

(2)

APPLICATION

4. Use RRAM with 4 equal subintervals to estimate the area between the curve $f(x) = \sqrt{x} + 4$ and the x-axis on the interval $3 \leq x \leq 5$, to the nearest tenth. Show your work.

(3)

5. Given the graph $f(x)$ as shown and the definition of $g(x) = \int_0^x f(t) \, dt$.

a. Find $g(-1)$, $g'(-1)$, $g''(-1)$.

(2)

b. Over what interval is $g(x)$ increasing? Show your work and explain your reasoning.

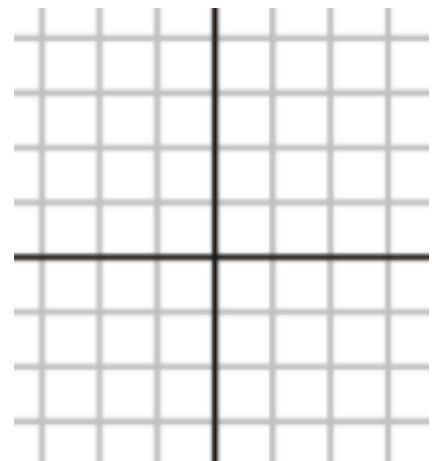
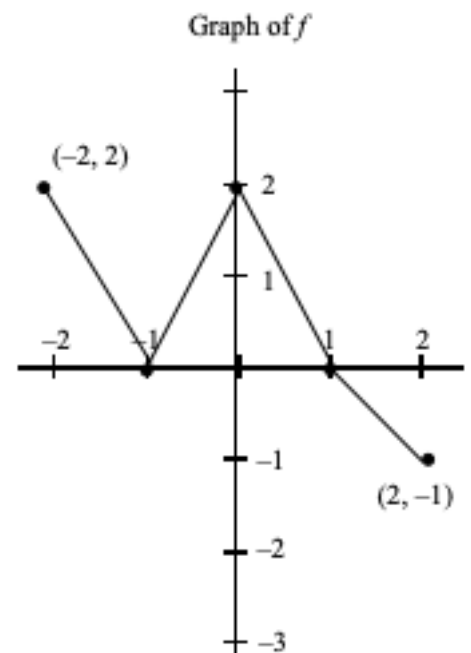
(1)

c. Over what interval is $g(x)$ concave up? Show your work and explain your reasoning.

(1)

d. Graph $g(x)$

(2)



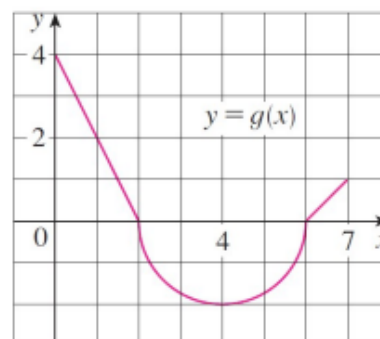
6. The temperature, in degrees Celsius ($^{\circ}\text{C}$), of the water in a pond is a differentiable function W of time t . The table below shows the water temperature as recorded every 3 days over a 15-day period. Approximate the average temperature, in degrees Celsius, of the water over the time interval $0 \leq t \leq 15$ days by using trapezoidal approximation with subintervals of length $\Delta t = 3$ days.

t (days)	$W(t)$ ($^{\circ}\text{C}$)
0	20
3	31
6	28
9	24
12	22
15	21

(3)

COMMUNICATION

7. The graph $y = g(x)$ shown on the right consists of two straight lines and a semicircle. Using the graph, **explain with detail** how you would evaluate $\int_0^7 g(x) dx$.



(2)

8. Helium is leaking from a large advertising balloon at a rate of $H(t)$ cubic centimeters per minute, where t is measured in minutes.

a. Write an integral that models the change in the amount of helium in the balloon over the interval $a \leq t \leq b$.

(1)

b. What are the units of the integral from part a?

(1)

c. Interpret $\int_0^{300} H(t) dt = -100$ in the context of the problem.

(1)

THINKING

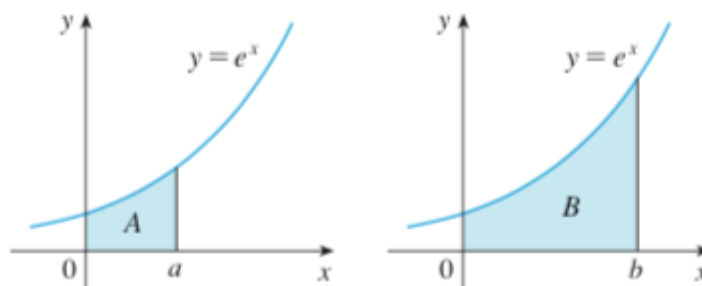
9. Determine the total area between the curve $f(x) = -6x^2 + 54$ and the x-axis on the interval $[1, 5]$. Show all work leading to the final answer.

(3)

10. Find a function f and a number a such that $6 + \int_0^x \frac{f(t)}{t^2} dt = 2\sqrt{x}$ for all $x > 0$.

(2)

11. The area labeled B is three times the area labeled A. Express b in terms of a .



(2)